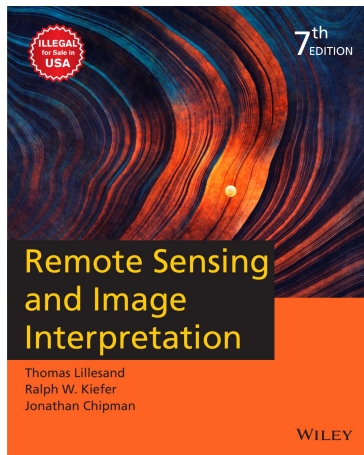


Satellite Imagery Classification

Lillesand, Chipman & Kiefer, 2015.

August 25, 2026

Remote Sensing and Image Interpretation



Content has been extracted from *Remote Sensing and Image Interpretation*, Seventh Edition. Wiley. By Thomas Lillesand, Jonathan Chipman & Ralph Kiefer, 2015. Visit <https://www.wiley.com/> for more info.

Outline

Image Classification

Supervised Classification

The Classification Stage

The Training Stage

Plan

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Spectral Pattern Recognition

- ▶ **The Core Objective:** Automatically categorize pixels into land cover classes or “themes”.
- ▶ **The “Individualist” Approach:**
 - ▶ Classification is performed **pixel-by-pixel**.
 - ▶ Based on spectral reflectance or emissivity signatures.
 - ▶ **Crucial Note:** It ignores the pixel’s neighbors or surroundings.
- ▶ **Extending the Model:**
 - ▶ **Polarization:** Radar imagery.
 - ▶ **Temporal:** Multitemporal sequences.
 - ▶ **Bidirectional:** Multi-angle imagery (e.g., MISR).

Spatial Pattern Recognition

- ▶ **The “Contextual” Approach:** Categorizes pixels based on their relationship with surrounding pixels.
- ▶ **Key Variables Considered:**
 - ▶ Texture, proximity, size, and shape.
 - ▶ Directionality, repetition, and context.
- ▶ **Human Synthesis:** Attempts to replicate the visual interpretation process of a human analyst.
- ▶ **Complexity:** More computationally intensive than spectral methods.
- ▶ **Hybrid Models (OBIA):** Object-Based Image Analysis combines both spectral and spatial logic.

Supervised Classification

The analyst acts as the teacher for the algorithm.

1. **Training Stage:** The analyst identifies representative sample sites of known cover types (**Training Areas**).
2. **Interpretation Key:** These areas are used to compile numerical descriptors of spectral attributes.
3. **Classification:** The computer compares every unknown pixel to the key.
4. **Labeling:** The pixel is assigned to the category it “looks most like” based on statistical strategies.

Unsupervised and Hybrid Classification

Unsupervised Classification

- ▶ **Data-First:** Pixels are grouped into “natural” clusters based on spectral similarity.
- ▶ **Post-Classification Identity:** The analyst labels the clusters *after* the grouping is done using ground reference data.

Hybrid & Specialized Procedures

- ▶ **Hybrid:** Combines supervised and unsupervised steps for better efficiency.
- ▶ **Advanced Topics:** Neural networks, mixed pixel analysis, and specialized hyperspectral procedures.

“There is no single ‘right’ manner... approach depends on data nature and intended application.”

Plan

Image Classification

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Data Input and Spectral Patterns

- ▶ **Sensor Agnostic:** Procedures apply to airborne multispectral data and satellite platforms (Landsat, SPOT, WorldView-2).
- ▶ **Multichannel Framework:** Analysis typically involves multiple spectral bands, for example:
 - ▶ Visible: Blue, Green, Red
 - ▶ Infrared: Near-Infrared (NIR), Thermal Infrared
- ▶ **Digital Numbers (DNs):** The sensor measures scene radiance recorded as DN's for each pixel across all bands.
- ▶ **Basis for Classification:** If spectral response patterns are sufficiently distinct for different terrain features, they form a “signature” for automated categorization.

The Three-Step Procedure

Supervised classification follows a structured three-stage workflow:

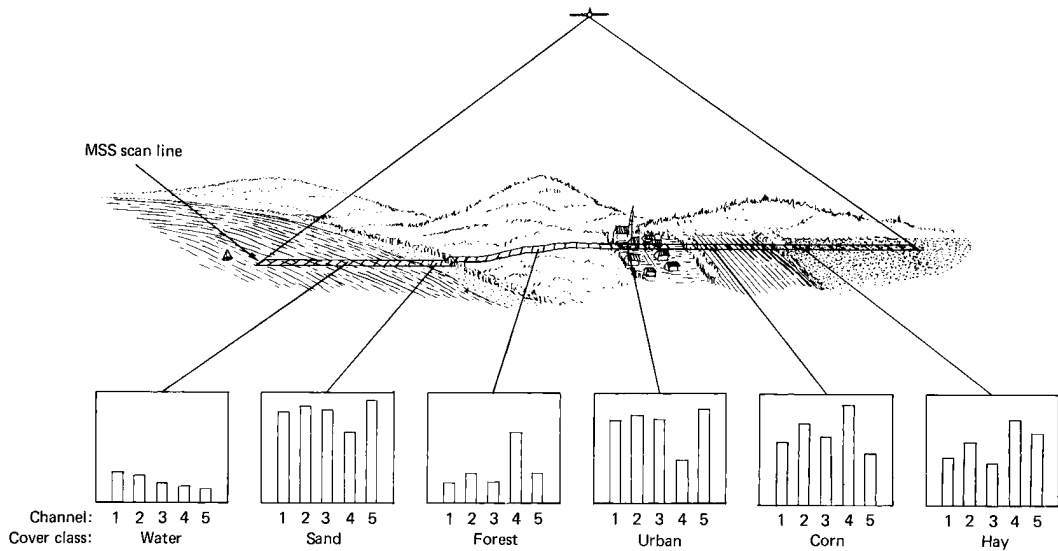
1. **Training Stage:** The analyst identifies representative “training areas” to develop numerical descriptions of the spectral attributes for each land cover type.
2. **Classification Stage:** Each pixel is compared to the training data and categorized into the class it most closely resembles.
3. **Output Stage:** Results are compiled into final products for end-users.

Pixels that do not sufficiently match any training set are labeled as “unknown”.

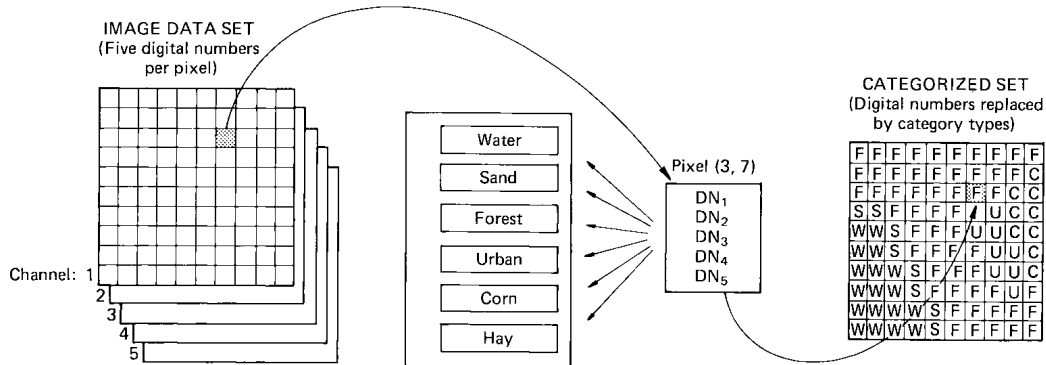
Classification Stage and Output Products

- ▶ **The “Heart” of the Process:** The classification stage uses computer-based **decision rules** to evaluate spectral patterns.
- ▶ **Versatile Output Formats:**
 - ▶ **Thematic Maps:** Visual representations of land cover.
 - ▶ **Statistical Tables:** Summary data for each class.
 - ▶ **Digital Data Files:** Classified data can be exported directly into a **Geographic Information System (GIS)**.
- ▶ **Strategic Integration:** Classification output often becomes GIS input, enabling complex spatial modeling and decision-making.

Multispectral Scan Line Profile



Basic steps



(1) **TRAINING STAGE**
Collect numerical data from training areas on spectral response patterns of land cover categories

(2) **CLASSIFICATION STAGE**
Compare each unknown pixel to spectral patterns; assign to most similar category

(3) **OUTPUT STAGE**
Present results:
maps
tables of area data
GIS data files

Plan

Image Classification

Supervised Classification

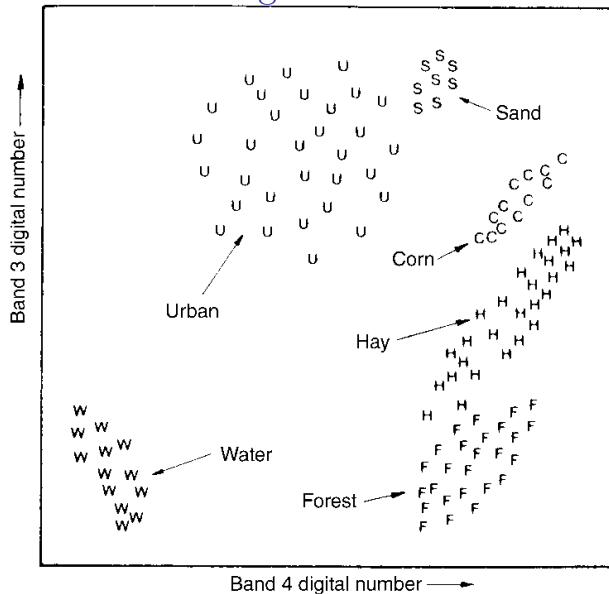
The Classification Stage

The Training Stage

The Concept of Measurement Space

- ▶ **Measurement Vectors:** Each pixel is represented as a point in a multi-dimensional space.
- ▶ **Scatter Plots:** For a 2-band subset, DNs from Band 3 (y -axis) and Band 4 (x -axis) define a pixel's coordinates.
- ▶ **Spectral Clusters:** Training data forms “clouds of points” rather than single values.
- ▶ **Natural Variability:** These clouds illustrate the centralizing tendency and inherent variance of specific land cover classes.

Pixel observations to scatter diagram



Minimum-Distance-to-Means Classifier

- ▶ **Mechanism:**

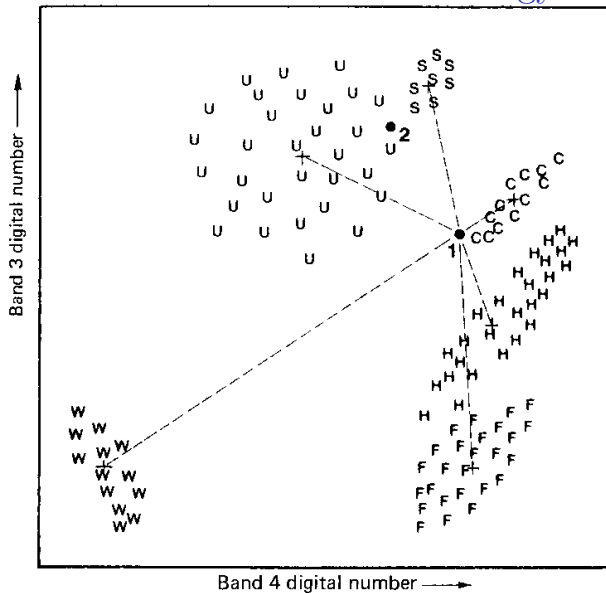
1. Calculate the **mean vector** (average spectral value) for each category.
2. Compute the Euclidean distance from an unknown pixel to each mean.
3. Assign pixel to the “closest” class.

- ▶ **Pros:** Mathematically simple and computationally efficient.

- ▶ **Cons:** Insensitive to different degrees of **variance**.

- ▶ **Limitation:** A pixel may be mathematically closer to a “tight” cluster (like sand) even if it logically belongs to a “wide” cluster (like urban).

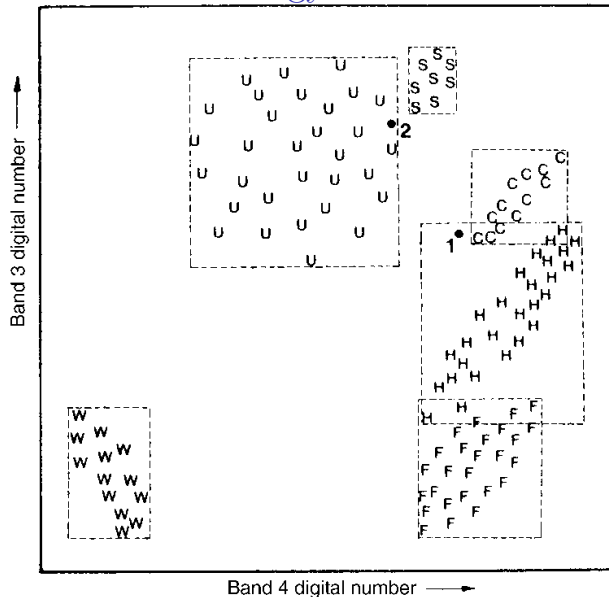
Minimum distance to means classification strategy



Parallelepiped Classifier

- ▶ **Mechanism:** Uses the range (min/max) of DN's in each band to create “decision regions” (rectangles in 2D, parallelepipeds in n-D).
- ▶ **Sensitivity:** Accounts for category variance by sizing the boxes differently.
- ▶ **The Overlap Problem:**
 - ▶ If categories overlap, pixels are labeled “not sure”.
 - ▶ Fails to account for **covariance** (slanted clouds where bands vary together).
- ▶ **Performance:** Very fast, but often poor for highly correlated spectral response patterns.

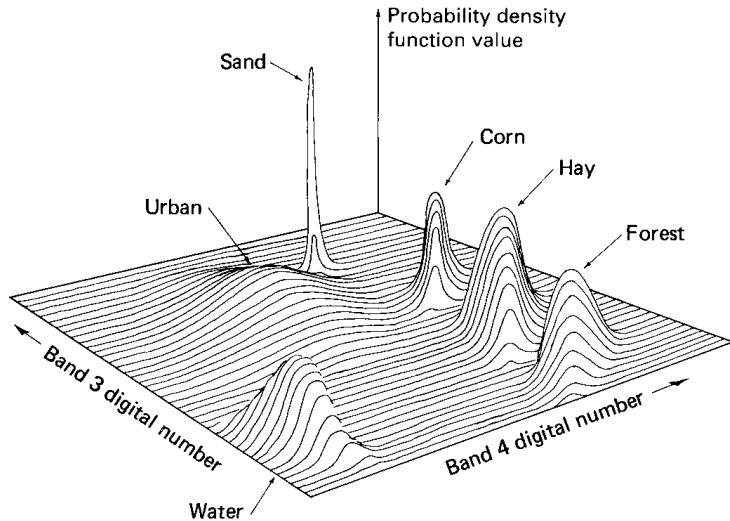
Parallelepiped classification strategy



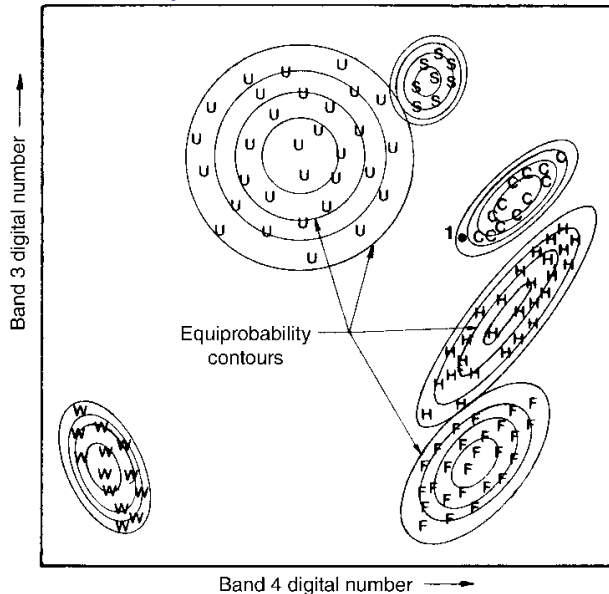
Gaussian Maximum Likelihood Classifier (GMLC)

- ▶ **The Probabilistic Leap:** Evaluates both variance and covariance by assuming a **Gaussian (Normal)** distribution.
- ▶ **Parameters:** Completely described by the mean vector μ and the covariance matrix Σ .
- ▶ **Mechanism:** Calculates the statistical probability of a pixel belonging to each class; assigns it to the “most likely” one.
- ▶ **Equi-probability Contours:** Delineates ellipsoidal decision regions that follow the “slant” of the data.

Probability density by a maximum likelihood classifier.



Equiprobability contours by a maximum likelihood classifier.



Bayesian Classifier & Cost Functions

The Bayesian approach refines GMLC by adding two weighting factors:

1. **A Priori Probability:** The anticipated likelihood of a class occurring in the scene (e.g., urban is more likely than rare sand).
2. **Cost of Misclassification:** A weight applied to minimize the “damage” of specific errors.

The Ideal Goal

To minimize the overall “cost” of errors, resulting in a theoretically optimum classification.

Efficiency vs. Accuracy

- ▶ **The Drawback of GMLC:** Large number of computations, especially with many bands.
- ▶ **Dimensionality Reduction:** Using Principal Component transformations to speed up the process.
- ▶ **Decision Tree (Layered) Classifiers:**
 - ▶ A “multi-step” approach.
 - ▶ Separate simple classes (like water) first using basic thresholds.
 - ▶ Reserve complex GMLC logic only for “ambiguous” overlapping classes.

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The Analyst's Critical Role

- ▶ **Manual vs. Automated:** While classification is automated, training is a manual process requiring “art and science”.
- ▶ **The Objective:** To define the location, size, and orientation of the “clouds of points” (spectral signatures) for each class.
- ▶ **Requirements:**
 - ▶ Close interaction with image data.
 - ▶ Substantial reference data.
 - ▶ Thorough knowledge of the geographic area.
- ▶ **Success Factor:** The quality of training determines the value of the entire classification effort.

Information Classes vs. Spectral Classes

- ▶ **Information Class:** The desired category (e.g., “Water” or “Agriculture”).
- ▶ **Spectral Class:** The distinct spectral subgroups within an information class.
- ▶ **The “Representative” Rule:**
 - ▶ One information class often requires **multiple** spectral classes.
 - ▶ **Example:** “Water” might include clear water, turbid water, and deep water (each requiring its own training set).
 - ▶ **Complexity:** Agriculture may vary by planting date, soil moisture, and crop variety.

Methods of Delineation

Manual Polygons:

- ▶ Analyst draws boundaries around known areas.
- ▶ **Edge Avoidance:** Avoid pixels on boundaries or “rough” areas to ensure purity.

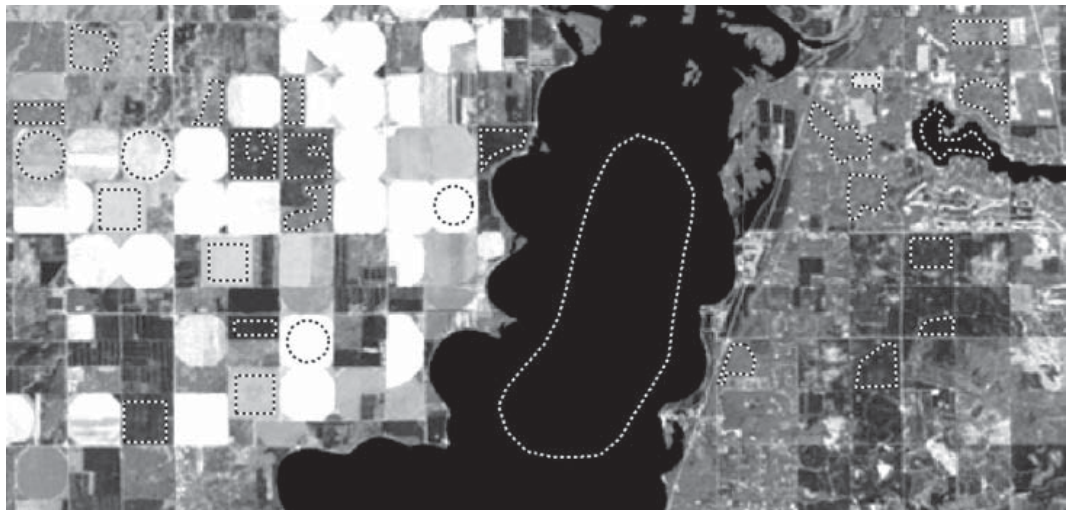
Seed Pixel Approach:

- ▶ Analyst selects a representative “seed”.
- ▶ Algorithm grows the area by including contiguous pixels with similar spectral traits.

Statistical Requirements

- ▶ **Sample Size (n = number of bands):**
 - ▶ Theoretical minimum: $n + 1$ pixels.
 - ▶ **Practical Rule:** Use $10n$ to $100n$ pixels per training set.
- ▶ **Spatial Dispersion:**
 - ▶ It is better to have many small sites (e.g., 20 sites of 40 pixels) than one giant site (1 site of 800 pixels).
 - ▶ Dispersion captures scene-wide variations.
- ▶ **Mathematical Basis:** These pixels generate the **mean vector** and **covariance matrix** for the classifier.

Training area polygons



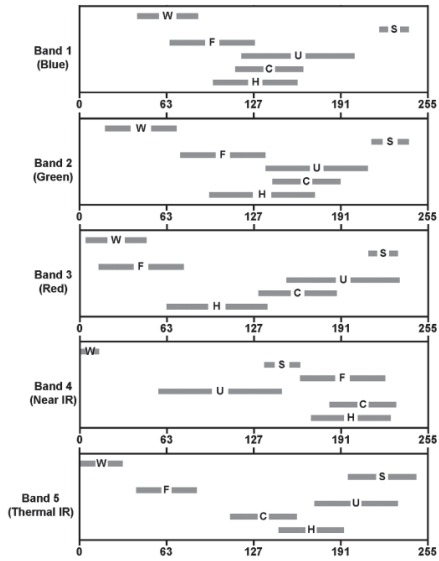
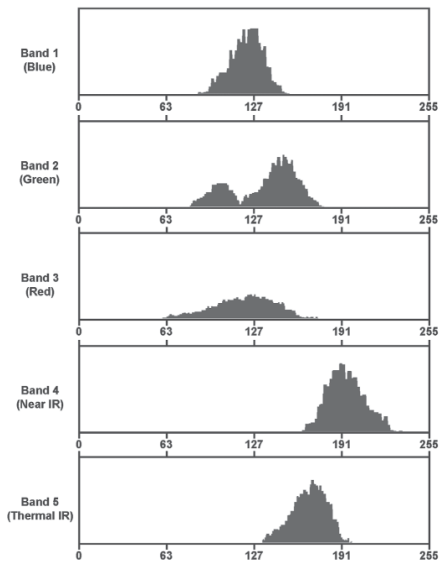
The Refinement Process

- ▶ **Statistical Testing:** Ensure data is **unimodal** and Gaussian (normally distributed).
- ▶ **Cleaning Data:**
 - ▶ Delete “edge pixels” or outliers (e.g., bare soil found within a crop field).
 - ▶ Split bimodal distributions into two distinct spectral classes.
- ▶ **Spectral Separability:** Evaluate if training sets for different classes overlap too much.
- ▶ **Final Goal:** A clean, non-redundant set of “interpretation keys” for the computer to follow.

Graphical Analysis: Histograms

- ▶ **Visual Normality Check:** Essential for Maximum Likelihood classifiers to ensure data follows a Gaussian distribution.
- ▶ **Identifying Subclasses:**
 - ▶ A **bimodal distribution** (two peaks) suggests the training site contains two distinct subclasses.
 - ▶ *Example:* “Hay” might actually be two different varieties or have different illumination conditions.
- ▶ **Action:** Split bimodal classes into separate spectral categories to improve overall accuracy.

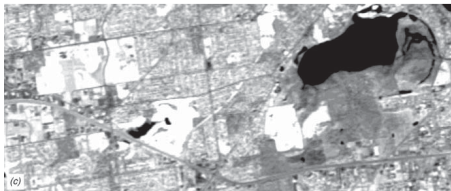
Visualization of training plot data



Coincident Spectral Plots

- ▶ **Comparison Tool:** Unlike histograms, these allow for direct comparison between different category types.
- ▶ **Visualization:** Displays mean response and variance (± 2 standard deviations) for each band.
- ▶ **Key Insight:** Reveals spectral overlap between classes (e.g., Hay vs. Corn).
- ▶ **Band Selection:** Helps identify which bands offer the best “reversals” or gaps for class separation.

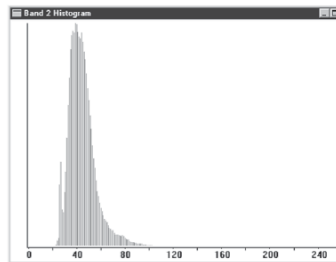
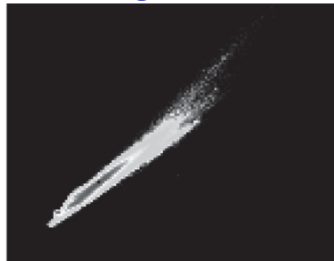
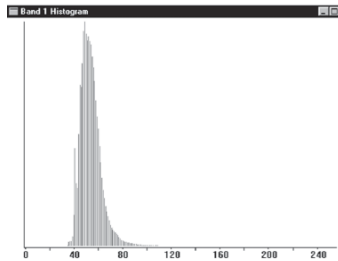
Multispectral Images



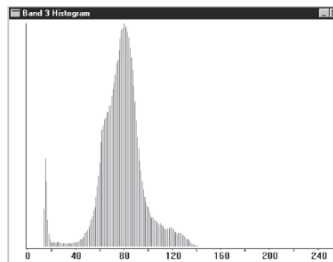
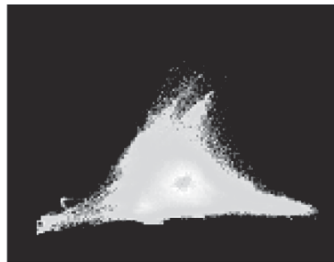
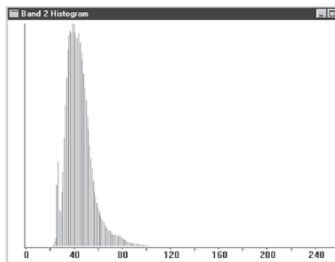
Scatter Diagrams (Measurement Space)

- ▶ **Multi-Band View:** Captures relationships that single-band histograms miss.
- ▶ **Correlation Analysis:**
 - ▶ **High Correlation:** Near-linear “cloud of points” (e.g., Visible Green vs. Red).
Harder to separate classes.
 - ▶ **Low Correlation:** Expanded “measurement space” (e.g., Red vs. Near-IR).
Much easier to separate land cover types.
- ▶ **Practical Application:** Often, just two well-chosen, low-correlation bands are enough for generalized classification.

Histograms and two-dimensional scatter diagram



Histograms and two-dimensional scatter diagram



Quantitative Separability Measures

- ▶ **Statistical Distance:** Computing a “matrix of divergence” between all class pairs.
- ▶ **Transformed Divergence (TD):** A covariance-weighted distance.
 - ▶ Values > 1500 generally indicate good separation.
 - ▶ Values < 1500 suggest problematic spectral overlap.
- ▶ **Jeffries-Matusita (JM) Distance:** Similar to TD but with a fixed scale (maximum 1414).
- ▶ **Diagnosis:** If overlap occurs between *different* information classes (e.g., Hay and Corn), refinement is needed.

Divergence Matrix

TABLE 7.1 Portion of a Divergence Matrix Used to Evaluate Pairwise Training Class Spectral Separability

Spectral Class ^a	W1	W2	W3	C1	C2	C3	C4	H1	H2...
W1	0								
W2	1185	0							
W3	1410	680	0						
C1	1997	2000	1910	0					
C2	1953	1890	1874	860	0				
C3	1980	1953	1930	1340	1353	0			
C4	1992	1997	2000	1700	1810	1749	0		
H1	2000	1839	1911	1410	1123	860	1712	0	
H2	1995	1967	1935	1563	1602	1197	1621	721	0
⋮	⋮								

^aW, water; C, corn; H, hay.

Self-Classification of Training Data

- ▶ **The “Preliminary Test”:** Classifying only the training pixels to see if the computer assigns them correctly.
- ▶ **The Error Matrix:** Displays percentages of “as expected” vs. “misclassified” training pixels.
- ▶ **The Warning:** Do NOT confuse this with final map accuracy.
 - ▶ Training areas are “pure” examples; the rest of the scene is usually much “messier”.
 - ▶ High training accuracy does not guarantee high scene-wide accuracy.

Interactive Preliminary Classification

- ▶ **Real-Time Feedback:** Using efficient algorithms (like parallelepiped) to highlight areas the current statistics would capture.
- ▶ **Visual Approximation:** Highlighting “classified” pixels in color over the raw imagery.
- ▶ **Representative Subsets:** Classifying a small, diverse section of the image first to verify the logic before committing to the full scene.
- ▶ **Iteration:** “Trial and error” testing of alternative deletions and poolings of training classes.

Tactics for Difficult Overlaps

When spectral classes refuse to separate, the analyst has several choices:

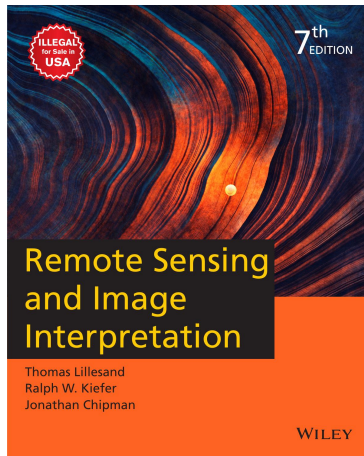
1. **Merger/Aggregation:** Combine specific classes into broader categories (e.g., Merge “Birch” and “Aspen” into “Deciduous”).
2. **Deletion:** Eliminate rare problem classes to preserve the accuracy of extensive, similar classes.
3. **Trial and Error:** Adjust sample sizes or re-evaluate the identity of training sites.

“If classes are inherently similar, no amount of retraining will make them separable!”

Multi-Image and Ancillary Data

- ▶ **The Multi-Image Problem:** Atmospheric and sun-angle changes mean “signatures” from one date rarely work on another.
- ▶ **Radiometric Normalization:** Matching secondary images to a “base” image to allow signature extension.
- ▶ **The Hybrid Future:**
 - ▶ Incorporating **GIS data** (elevation, soil type).
 - ▶ Using **Multitemporal** data (seasonal changes).
 - ▶ **Spatial Pattern Recognition** to mimic human visual interpretation.

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